Waves

90 questions: 45 multiple choice, 45 structured.

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1 · Part 1 Core General properties of waves [1]

A stone is dropped into a still pond. Ripples spread out across the surface of the water. A small leaf is floating on the water some distance away.

What happens to the leaf as the ripples pass it?

A It is carried outwards with the ripples to the edge of the pond.

B It moves up and down but stays in about the same place.

C It stays completely still.

D It is pulled towards the point where the stone landed.

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2 · Part 1 Core General properties of waves [1]

Which pair contains only waves that can be modelled as longitudinal waves?

A sound waves and seismic P-waves

B sound waves and seismic S-waves

C light waves and water waves

D water waves and seismic P-waves

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3 · Part 1 Core General properties of waves [1]

A student moves one end of a long rope up and down 4.0 times each second. This sends a wave along the rope. The wavelength of the wave is 0.75 m.

What is the speed of the wave along the rope?

A 0.19 m / s

B 4.8 m / s

C 5.3 m / s

D 3.0 m / s

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4 · Part 1 Core General properties of waves [1]

A wave travels across the surface of a lake. The vertical distance from the top of a crest to the bottom of a trough is 12 cm. The horizontal distance from a crest to the next trough is 40 cm.

What are the amplitude and the wavelength of the wave?

	amplitude / cm	**wavelength / cm**
A	6	40
B	6	80
C	12	40
D	12	80

5 · Part 1 Core General properties of waves [1]

In a ripple tank, water waves travel from a region of deep water into a region of shallow water.

What happens to the speed, the wavelength and the frequency of the waves?

	speed	**wavelength**	**frequency**
A	increases	increases	stays the same
B	decreases	stays the same	decreases
C	decreases	decreases	stays the same
D	stays the same	decreases	increases

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6 · Part 1 Supplement General properties of waves [1]

Straight water waves in a ripple tank pass through a gap between two barriers. The wavelength of the waves and the width of the gap are changed.

Which combination makes the waves spread out the most after they pass through the gap?

	wavelength / cm	gap width / cm
A	2.0	2.0
B	1.0	5.0
C	2.0	6.0
D	0.5	5.0

7 · Part 1 Supplement General properties of waves [1]

Straight sea waves travel past the end of a long harbour wall. Some of the waves spread into the calm water behind the wall.

Which change makes the waves spread further into the region behind the wall?

A waves of larger amplitude

B waves of shorter wavelength

C a longer harbour wall

D waves of longer wavelength

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8 · Part 1 Core Reflection of light [1]

A ray of light strikes a plane mirror. The angle between the incident ray and the surface of the mirror is 35°.

What is the angle of reflection?

A 35°

B 70°

C 55°

D 110°

9 · Part 1 Core Reflection of light [1]

A boy stands 1.5 m in front of a vertical plane mirror.

Which statement describes his image?

A It is real and 1.5 m behind the mirror.

B It is virtual and 1.5 m behind the mirror.

C It is virtual and 3.0 m behind the mirror.

D It is virtual and on the surface of the mirror.

10 · Part 1 Supplement Reflection of light [1]

A ray of light strikes a plane mirror with an angle of incidence of 25°. The incident ray is kept fixed in the same position. The mirror is turned so that the angle of incidence becomes 35°.

Through what angle does the reflected ray turn?

A 10°

B 35°

C 70°

D 20°

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11 · Part 1 Core Refraction of light [1]

A ray of light in air strikes the flat surface of a glass block. The angle of incidence is 40°.

Which statement describes the ray inside the glass?

A It bends towards the normal and the angle of refraction is less than 40°.

B It bends away from the normal and the angle of refraction is more than 40°.

C It does not change direction.

D It is totally internally reflected at the surface.

12 · Part 1 Core Refraction of light [1]

What is meant by the critical angle?

A the angle of incidence in air at which light is totally internally reflected

B the angle of refraction in glass when the angle of incidence in air is 0°

C the angle of incidence in the optically denser material for which the angle of refraction is 90°

D the angle between the ray and the boundary at which all of the light is reflected

13 · Part 1 Core Refraction of light [1]

Which of these is an example of total internal reflection?

A a swimming pool appearing shallower than it really is

B an image seen in a flat bathroom mirror

C a straw appearing bent where it enters a glass of water

D light being turned through 90° inside a glass prism in a periscope

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14 · Part 1 Supplement Refraction of light [1]

A ray of light passes from air into a plastic block. The angle of incidence is 45° and the angle of refraction is 28°.

What is the refractive index of the plastic?

A 0.66

B 1.5

C 1.6

D 0.62

15 · Part 1 Supplement Refraction of light [1]

A ray of light travels inside a block of glass of refractive index 1.60. It strikes the boundary between the glass and the air with an angle of incidence of 42°.

What happens to the ray at the boundary?

A It passes into the air with an angle of refraction of 25°.

B It travels along the boundary with an angle of refraction of 90°.

C It is totally internally reflected with an angle of reflection of 42°.

D It is totally internally reflected with an angle of reflection of 39°.

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16 · Part 1 Core General properties of waves [5]

A long spring is stretched along a bench. A student repeatedly pushes and pulls one end of the spring backwards and forwards along the length of the spring. A wave travels along the spring. A small piece of tape is stuck to one coil near the middle of the spring.

(a) Name the type of wave produced on the spring. [1]

(b) Describe the motion of the piece of tape as the wave passes it. [2]

(c) State what the wave transfers along the spring. [1]

(d) The student now moves the end of the spring from side to side across the bench, at right angles to the length of the spring. A different type of wave travels along the spring.

State the direction in which the coils vibrate compared with the direction in which this wave travels. [1]

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17 · Part 1 Core General properties of waves [6]

A fisherman watches waves on the sea pass a wooden post at the end of a jetty. He counts 12 complete waves passing the post in 60 s. The distance between neighbouring crests is 7.5 m.

(a) State what is meant by the frequency of a wave. [1]

(b) Calculate the frequency of the waves. [2]

(c) Calculate the speed of the waves. [2]

(d) Later, waves travel at the same speed as in (c) but have a frequency of 0.25 Hz.

Determine the wavelength of these waves. [1]

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18 · Part 1 Core General properties of waves [6]

An earthquake produces seismic P-waves and seismic S-waves. Both types of wave travel through the Earth to a recording station 360 km away. The P-waves travel at 8000 m / s and the S-waves travel at 4500 m / s.

(a) State which of the two types of seismic wave can be modelled as a transverse wave. [1]

(b) Calculate the time between the arrival of the P-waves and the arrival of the S-waves at the recording station. [3]

(c) The S-waves have a frequency of 0.50 Hz.

Calculate the wavelength of the S-waves. Give your answer in metres. [2]

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19 · Part 1 Core General properties of waves [6]

A student uses a ripple tank to show refraction of water waves. A vibrating straight bar makes straight waves in the deep part of the tank.

(a) Describe how the student can set up the ripple tank so that the waves travel from deep water into shallow water, and state what she observes when the waves meet the boundary at an angle. [2]

(b) In the deep water the waves travel at 24 cm / s and have a wavelength of 1.2 cm.

Calculate the frequency of the waves. [2]

(c) In the shallow water the wavelength of the waves is 0.80 cm.

Calculate the speed of the waves in the shallow water. [2]

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20 · Part 1 Core General properties of waves [5]

A ripple tank is used to show diffraction of water waves.

(a) Describe how the ripple tank is used to show diffraction of straight waves through a gap. [3]

(b) State what happens to the wavelength of the waves as they pass through the gap. [1]

(c) One of the barriers is removed, so that the waves now pass the edge of a single barrier.

Describe what happens to the waves as they pass the edge of the barrier. [1]

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21 · Part 1 Supplement General properties of waves [7]

A teacher is talking in a corridor, just to one side of an open doorway. A student in the classroom can hear the teacher clearly through the doorway but cannot see her. The doorway is 0.90 m wide.

The sound of the teacher's voice has a frequency of 380 Hz. The speed of sound in the air is 342 m / s. The wavelength of visible light is about 5.0×10^{-7} m.

(a) Calculate the wavelength of the sound waves. [2]

(b) Explain why the sound waves spread out a lot after passing through the doorway but the light waves do not. [3]

(c) The teacher plays two notes on a loudspeaker, one of low frequency and one of high frequency. The sound waves pass the edge of the door frame.

State which note spreads further round the edge into the classroom, and explain why. [2]

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22 · Part 1 Core Reflection of light [6]

A girl uses a vertical plane mirror in a dance studio.

(a) Define the term normal, as used for the reflection of light. [1]

(b) A ray of light from a lamp strikes the mirror. The angle between the incident ray and the reflected ray is 64°.

Determine the angle of incidence. [2]

(c) The girl stands 2.5 m in front of the mirror. Her image is 2.5 m behind the mirror.

State two other characteristics of her image. [2]

(d) State the distance between the girl and her image. [1]

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23 · Part 1 Supplement Reflection of light [5]

(a) In a simple periscope, a ray of light travelling horizontally strikes a plane mirror. The surface of the mirror makes an angle of 45° with the horizontal.

(i) State the angle of incidence of the ray at the mirror. [1]

(ii) State the angle between the incident ray and the reflected ray. [1]

(b) A man stands 4.0 m in front of a vertical plane mirror. He walks directly towards the mirror at a constant speed of 1.2 m / s for 2.5 s, then stops.

Calculate the distance between the man and his image when he stops. [3]

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24 · Part 1 Core Refraction of light [6]

A student is given a rectangular glass block, a ray box, a sheet of paper, a pencil, a ruler and a protractor.

(a) Describe how the student can use this apparatus to trace the path of a ray of light through the block and measure the angle of incidence and the angle of refraction where the ray enters the block. [4]

(b) The ray leaves the rectangular block through the face opposite the face where it entered.

State how the direction of the ray that leaves the block compares with the direction of the ray that enters the block. [1]

(c) The student then uses a semicircular glass block. She aims a ray at the centre of the flat face, so that the ray enters the block through the curved face along a radius.

State why the ray does not change direction as it enters the curved face. [1]

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25 · Part 1 Core Refraction of light [5]

(a) Define the term angle of refraction. [1]

(b) A ray of light in air travels straight down into a still pond, along the normal to the water surface.

State the direction of the ray in the water. [1]

(c) A ray of light travels inside a glass window towards its surface and passes out into the air. Its angle of incidence at the glass surface is 25°, which is less than the critical angle.

State whether the angle of refraction in the air is greater than or less than 25°, and explain your answer. [2]

(d) State what happens to the speed of light as it passes from air into glass. [1]

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26 · Part 1 Core Refraction of light [6]

A student aims a ray of light through the curved face of a semicircular glass block towards the centre of its flat face. Inside the glass, the ray strikes the flat face, where the glass meets the air. The student slowly increases the angle of incidence at the flat face from 0° to 80°.

(a) State the meaning of the term critical angle. [1]

(b) Describe what the student observes at the flat face as the angle of incidence increases from 0° to 80°. [3]

(c) State the two conditions needed for total internal reflection to happen. [2]

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27 · Part 1 Supplement Refraction of light [6]

Sea waves travel towards a beach. They pass from deeper water into shallower water across a straight boundary. The waves travel at 6.0 m / s in the deeper water and at 4.0 m / s in the shallower water.

(a) Define refractive index, n. [1]

(b) Calculate the refractive index for the waves passing from the deeper water into the shallower water. [2]

(c) The waves meet the boundary with an angle of incidence of 40°.

Calculate the angle of refraction of the waves in the shallower water. [3]

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28 · Part 1 Supplement Refraction of light [8]

A teacher demonstrates how an optical fibre works using a long, thin, solid glass rod surrounded by air. The refractive index of the glass is 1.48.

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(a) Describe how light travels along the glass rod from one end to the other. [2]

(b) Calculate the critical angle for light in the glass meeting the glass–air boundary. [2]

(c) A ray of light inside the rod strikes the side of the rod with an angle of incidence of 40°.

State and explain what happens to this ray at the side of the rod. [2]

(d) Optical fibres are widely used in telecommunications.

Give two reasons why optical fibres are used to carry telephone and internet signals. [2]

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29 · Part 1 Supplement Refraction of light [5]

A student shines rays of light from air into a rectangular block of acrylic plastic. She measures the angle of refraction for three different angles of incidence. Her results are shown in the table.

angle of incidence / °	angle of refraction / °
20	13
40	25
60	35

(a) Use the result for an angle of incidence of 40° to calculate the refractive index of the acrylic. [2]

(b) Predict the angle of refraction when the angle of incidence is 50°. [2]

(c) State the angle of refraction when the ray enters the block along the normal. [1]

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30 · Part 1 Core Refraction of light [6]

A glass prism used in a pair of binoculars has a cross-section that is a right-angled triangle. Its other two angles are both 45°. A ray of light in air enters the prism at right angles to one of the two shorter faces. Inside the glass, the ray strikes the longest face with an angle of incidence of 45°. The critical angle of the glass is 42°.

(a) State why the ray does not change direction as it enters the prism. [1]

(b) Explain why the ray is totally internally reflected at the longest face. [2]

(c) State the angle through which the ray has been turned after it reflects from the longest face. [1]

(d) The glass prism is replaced by a prism of the same shape made from a plastic with a critical angle of 48°. The ray enters in the same way.

Describe what happens to the ray when it strikes the longest face. [2]

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1 · Part 2 Core Thin lenses [1]

Parallel rays of light, travelling parallel to the principal axis, pass through a thin diverging lens.

Which statement describes the light after it leaves the lens?

A The rays meet at a single point on the far side of the lens.

B The rays spread out as if they came from a point on the same side of the lens as the incoming light.

C The rays stay parallel to each other but move closer together.

D The rays spread out as if they came from a point on the far side of the lens.

2 · Part 2 Core Thin lenses [1]

Which statement describes the principal focus of a thin converging lens?

A the centre of the lens, where rays pass through without changing direction

B the point where an image is formed, whatever the position of the object

C the point on the principal axis at a distance of twice the focal length from the lens

D the point where rays parallel to the principal axis meet after passing through the lens

3 · Part 2 Core Thin lenses [1]

The flame of a candle is placed on the principal axis of a thin converging lens of focal length 10 cm. The flame is 30 cm from the lens. A sharp image of the flame forms on a screen on the other side of the lens.

Which describes the image?

A real, inverted, diminished

B real, inverted, enlarged

C virtual, upright, enlarged

D real, upright, diminished

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4 · Part 2 Core Thin lenses [1]

Which statement about a virtual image is correct?

A It is formed at the point where rays of light actually meet.

B It can be seen on a screen placed at the position of the image.

C It is located by extending diverging rays backwards, and it cannot be formed on a screen.

D It is formed only when the object is more than two focal lengths from a lens.

5 · Part 2 Supplement Thin lenses [1]

Mei is short-sighted.

Which statement describes her problem and how it is corrected?

A Light from distant objects is focused in front of the retina. It is corrected with a diverging lens.

B Light from distant objects is focused in front of the retina. It is corrected with a converging lens.

C Light from near objects is focused behind the retina. It is corrected with a diverging lens.

D Light from near objects is focused behind the retina. It is corrected with a converging lens.

6 · Part 2 Core Dispersion of light [1]

White light in a narrow beam passes through a glass prism. A spectrum of colours is seen on a white screen.

Which colour has been refracted by the greatest amount?

A red

B violet

C green

D yellow

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7 · Part 2 Core Dispersion of light [1]

Which list shows four colours of the visible spectrum in order of increasing wavelength?

A red, yellow, green, blue

B indigo, violet, blue, green

C green, blue, yellow, orange

D blue, green, yellow, orange

8 · Part 2 Supplement Dispersion of light [1]

What is meant by monochromatic light?

A light that contains all seven colours of the visible spectrum

B light that has passed through a glass prism

C visible light of a single frequency

D light that cannot be refracted by glass

9 · Part 2 Core Electromagnetic spectrum [1]

Which region of the electromagnetic spectrum has a longer wavelength than visible light but a shorter wavelength than microwaves?

A infrared

B ultraviolet

C radio waves

D X-rays

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10 · Part 2 Core Electromagnetic spectrum [1]

Radio waves and X-rays both travel through a vacuum.

Which statement is correct?

A X-rays travel faster because they have a higher frequency.

B Radio waves travel faster because they have a longer wavelength.

C X-rays travel faster because they can pass through more materials.

D Both travel at the same speed.

11 · Part 2 Core Electromagnetic spectrum [1]

Which row correctly matches a use with the region of the electromagnetic spectrum used?

	use	region
A	detecting fake bank notes	X-rays
B	thermal imaging	infrared
C	satellite television	gamma rays
D	mobile phones	ultraviolet

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12 · Part 2 Core Electromagnetic spectrum [1]

Excessive exposure to which type of radiation is most likely to cause mutation of cells in the body?

A infrared

B microwaves

C gamma rays

D visible light

13 · Part 2 Core Electromagnetic spectrum [1]

Direct broadcast television is sent to homes by a company using an artificial satellite.

Which row shows the type of satellite usually used and the region of the electromagnetic spectrum mainly used to communicate with it?

	type of satellite	region used
A	low orbit	radio waves
B	low orbit	microwaves
C	geostationary	radio waves
D	geostationary	microwaves

14 · Part 2 Supplement Electromagnetic spectrum [1]

Mobile phones use microwaves to send and receive signals.

Which statement gives reasons why microwaves are suitable?

A Microwaves can penetrate some walls and only need a short aerial.

B Microwaves pass through all walls without being weakened, but they need a very long aerial.

C Microwaves are reflected by every wall, so they reach every room of a building.

D Microwaves travel faster than radio waves, so the signal arrives sooner.

15 · Part 2 Supplement Electromagnetic spectrum [1]

Radio waves of frequency 120 MHz are emitted by a transmitter into the air.

What is the wavelength of these radio waves in air?

A 0.40 m

B 2500 m

C 2.5 m

D 3.6×10^{16} m

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16 · Part 2 Core Thin lenses [5]

A student shines a wide parallel beam of light onto a thin converging lens. The beam travels parallel to the principal axis of the lens. She moves a screen behind the lens until the light makes the smallest, brightest spot. The spot is 15 cm from the centre of the lens.

(a) Describe what happens to the rays of light as they pass through the converging lens. [2]

(b) State the name of the point where the smallest, brightest spot forms. [1]

(c) State the focal length of the lens. [1]

(d) The student replaces the converging lens with a thin diverging lens. State what happens to the parallel beam after it passes through the diverging lens. [1]

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17 · Part 2 Core Thin lenses [6]

A thin converging lens has a focal length of 8.0 cm. It is used to form an image of a small lamp filament on a screen.

(a) Define the focal length of a lens. [1]

(b) The filament is placed on the principal axis, 12 cm from the lens. A sharp image forms on the screen. Describe this image using three of the terms: enlarged, same size, diminished, upright, inverted, real, virtual. [3]

(c) The filament is moved so that it is 16 cm from the lens, and the screen is moved until the image is sharp again. State how the size of the image now compares with the size of the filament. [1]

(d) Give one reason why the images in (b) and (c) are described as real. [1]

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18 · Part 2 Supplement Thin lenses [6]

A jeweller uses a single thin converging lens of focal length 5.0 cm as a magnifying glass to examine a small gemstone. She looks at the gemstone through the lens.

(a) State where the gemstone must be placed, compared with the principal focus, for the lens to act as a magnifying glass. [1]

(b) Describe the image of the gemstone that the jeweller sees. Use three terms from: enlarged, same size, diminished, upright, inverted, real, virtual. [3]

(c) Explain why this image cannot be formed on a screen. [2]

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19 · Part 2 Supplement Thin lenses [6]

A man can see distant buildings clearly, but the words in a book held close to his face look blurred.

(a) State the name of this eye defect. [1]

(b) Light from the book enters his uncorrected eye. State where this light is brought to a focus. [1]

(c) State the type of lens used in spectacles to correct this defect. [1]

(d) Explain how this lens allows him to see the book clearly. [2]

(e) His daughter is short-sighted. State how the effect of the lens in her spectacles on light differs from the effect of the lens in his spectacles. [1]

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20 · Part 2 Core Dispersion of light [6]

A narrow beam of white light enters one face of a triangular glass prism. The light leaves through a second face and falls on a white screen, where a band of colours is seen.

(a) State the name of the process that splits white light into colours. [1]

(b) Explain why the white light separates into different colours in the prism. [2]

(c) State the colour that is refracted the least. [1]

(d) List the seven traditional colours of the visible spectrum in order of increasing frequency. [2]

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21 · Part 2 Supplement Dispersion of light [5]

A laser pointer produces a narrow beam of green light. The light from the laser is monochromatic.

(a) State what is meant by monochromatic. [1]

(b) The laser beam is passed through a glass prism and falls on a white screen. Predict what is seen on the screen. Explain your answer. [2]

(c) A beam of red light and a beam of violet light strike the same face of a glass prism at the same angle of incidence.

(i) State which beam changes direction by the greater amount. [1]

(ii) State which beam has the longer wavelength. [1]

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22 · Part 2

Core

Electromagnetic spectrum

[5]

The table lists the main regions of the electromagnetic spectrum in order of increasing frequency. Two regions are labelled P and Q.

region 1	region 2	region 3	region 4	region 5	region 6	region 7
radio waves	microwaves	P	visible light	ultraviolet	Q	gamma rays

(a) Name the regions P and Q. [2]

(b) State which region in the table has the longest wavelength. [1]

(c) State how the speed of gamma rays in a vacuum compares with the speed of radio waves in a vacuum. [1]

(d) State which region in the table can be detected by the human eye. [1]

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23 · Part 2 Core Electromagnetic spectrum [6]

A hospital uses several regions of the electromagnetic spectrum.

(a) State the region of the electromagnetic spectrum used for each of these.

(i) producing images of broken bones [1]

(ii) sterilising surgical instruments that are already sealed in plastic packets [1]

(iii) a camera that shows which patients have a high body temperature [1]

(iv) checking for fake bank notes in the hospital shop [1]

(b) The person who operates the machine in (a)(i) stands behind a protective screen each time a patient is scanned. Explain why. [2]

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24 · Part 2 Core Electromagnetic spectrum [5]

People can be harmed by excessive exposure to some types of electromagnetic radiation.

(a) A person spends many hours in strong sunshine without protection. Describe the harmful effects of excessive exposure to the ultraviolet in sunlight. [2]

(b) A student says: "Only radiation with a higher frequency than visible light can harm people."

Comment on this statement. In your answer, name two regions with a lower frequency than visible light and state the harmful effect of excessive exposure to each. [3]

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25 · Part 2 Core Electromagnetic spectrum [5]

A walker in a remote mountain area uses a satellite phone. Her family at home watches television received directly from a satellite.

(a) State the region of the electromagnetic spectrum mainly used for communication with artificial satellites. [1]

(b) State the type of satellite used for direct broadcast satellite television. [1]

(c) State two uses of the region you named in (a) that do not involve satellites. [2]

(d) State one use of radio waves. [1]

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26 · Part 2 Supplement Electromagnetic spectrum [6]

A kitchen oven heats food using microwaves of frequency 2.5 GHz.

(a) State the speed of microwaves in air. [1]

(b) Calculate the wavelength of these microwaves in air. [3]

(c) A local radio station transmits radio waves with a wavelength of 3.2 m in air. Calculate the frequency of these radio waves. [2]

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27 · Part 2 Supplement Electromagnetic spectrum [6]

A family home has a wireless internet router, a pair of Bluetooth headphones and a high-speed broadband connection through an optical fibre.

(a) State the region of the electromagnetic spectrum used for wireless internet. Give two reasons why this region is suitable. [3]

(b) The Bluetooth headphones use radio waves. The sound becomes less clear when the user walks into another room, away from the device sending the signal. Explain why. [1]

(c) State the types of electromagnetic radiation that can be used in the optical fibre. Give one reason why optical fibres are used for high-speed broadband. [2]

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28 · Part 2 Supplement Electromagnetic spectrum [5]

A singer's voice is recorded and sent to a studio as an electrical signal.

(a) State the difference between an analogue signal and a digital signal. [2]

(b) Two signals, X and Y, are described below.

Signal X: the voltage from the microphone changes smoothly and continuously, and can take any value between −0.2 V and +0.2 V.

Signal Y: the voltage was measured every millisecond, as shown in the table.

time / ms	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
voltage / V	0	5	5	0	5	0	0	5

The voltage of signal Y between readings is the same as at the reading before.

State whether each signal is analogue or digital. [2]

(c) A student says: "Sound can only be sent as an analogue signal, because a sound wave changes smoothly." State whether the student is correct. Give a reason. [1]

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29 · Part 2 Supplement Electromagnetic spectrum [6]

A telephone company sends signals along a cable hundreds of kilometres long. As a signal travels, it becomes weaker and picks up unwanted random changes called noise. At intervals along the cable, the signal is regenerated before it is sent on.

(a) State two benefits of digital signalling compared with analogue signalling. [2]

(b) Explain why a digital signal can be regenerated accurately but an analogue signal cannot. [3]

(c) Explain how accurate regeneration allows a digital signal to be sent over a greater distance. [1]

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30 · Part 2 Core Electromagnetic spectrum [6]

A distant star suddenly explodes. It gives out radio waves, visible light and gamma rays at the same instant. Detectors on a satellite in space record all three. Assume that all three types of radiation travel the whole way to the satellite through a vacuum.

(a) Write the three types of radiation in order of increasing wavelength. [1]

(b) Predict the order in which the three types of radiation reach the satellite. Explain your answer. [2]

(c) Gamma rays are also used on Earth. State two uses of gamma rays. [2]

(d) Another detector on the satellite measures ultraviolet. State one use of ultraviolet on Earth. [1]

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1 · Part 3 Core Sound [1]

A guitar string is plucked and a note is heard.

Which statement describes how the string produces the sound?

A The string gives out tiny sound particles that travel to the listener's ear.

B The string vibrates and makes the air particles next to it vibrate.

C The string warms the air next to it, and this warm air moves to the listener.

D The string pushes the air steadily away from it in one direction.

2 · Part 3 Core Sound [1]

Sound waves are longitudinal waves.

What does this mean?

A Sound waves can only travel over long distances.

B The particles of the medium vibrate at right angles to the direction in which the wave travels.

C The particles of the medium move along with the wave from the source to the listener.

D The particles of the medium vibrate parallel to the direction in which the wave travels.

3 · Part 3 Core Sound [1]

A signal generator and loudspeaker produce four sounds, one at a time.

Which sound can be heard by a young person with normal hearing?

A 12 000 Hz

B 8 Hz

C 35 kHz

D 150 000 Hz

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4 · Part 3 Core Sound [1]

Two astronauts are working in space outside their spacecraft. They are a few metres apart and are not touching each other or the spacecraft. One astronaut bangs two metal tools together. The other astronaut sees this happen but hears nothing.

Why does the second astronaut hear nothing?

A The sound has a frequency that is too high to hear.

B Sound travels too slowly in space to reach the second astronaut.

C There is no medium in space to transmit the sound.

D The helmet of the second astronaut reflects all of the sound.

5 · Part 3 Core Sound [1]

Which value is closest to the speed of sound in air?

A 340 m / s

B 34 m / s

C 3400 m / s

D 3.0×10^8 m / s

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6 · Part 3 Core Sound [1]

A student plays a note on a recorder. She then plays a second note that is louder and has a higher pitch than the first note.

Compared with the first note, how do the amplitude and the frequency of the second note's sound wave change?

	amplitude	**frequency**
A	larger	lower
B	larger	higher
C	smaller	higher
D	the same	higher

7 · Part 3 Core Sound [1]

A microphone is connected to an oscilloscope. Four different sounds are played into the microphone, one at a time. The oscilloscope settings are not changed.

For each sound, a student records the number of complete waves shown across the screen and the height of the peaks of the trace above the centre line.

Which sound has the highest pitch and is the quietest?

	number of complete waves on screen	**peak height of trace / cm**
A	2	3.0
B	6	3.0
C	2	1.0
D	6	1.0

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8 · Part 3 Core Sound [1]

A hiker in a valley shouts once towards a large, flat rock face. She hears the echo 0.50 s after she shouts.

The speed of sound in air is 340 m / s.

How far is the hiker from the rock face?

A 680 m

B 170 m

C 85 m

D 43 m

9 · Part 3 Core Sound [1]

Which statement about an echo is correct?

A An echo is sound that has been refracted by a wall.

B An echo is sound that has been reflected from a surface.

C An echo has a higher frequency than the original sound.

D An echo travels faster than the original sound.

10 · Part 3 Core Sound [1]

What is ultrasound?

A sound with a frequency higher than 20 kHz

B sound with a frequency lower than 20 Hz

C sound with a frequency higher than 20 Hz

D an electromagnetic wave with a frequency higher than 20 kHz

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11 · Part 3 Core Sound [1]

Student X hits a metal dustbin lid with a hammer. Student Y stands 660 m away in a large, flat park. Student Y starts a stopwatch when she sees the hammer hit the lid and stops it when she hears the sound. The stopwatch reads 2.0 s.

What value for the speed of sound do these results give?

A 1320 m / s

B 165 m / s

C 0.0030 m / s

D 330 m / s

12 · Part 3 Supplement Sound [1]

A sound wave travels through air.

What is a rarefaction?

A a region where the air particles are closer together than normal

B a region where the air particles are not vibrating

C a region where the air particles are further apart than normal

D a region where the sound wave is reflected

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13 · Part 3 Supplement Sound [1]

The table gives the speed of sound in three materials X, Y and Z. One material is air, one is water and one is steel.

material	speed of sound / m / s
X	1480
Y	340
Z	5100

Which row identifies the three materials?

	X	Y	Z
A	air	water	steel
B	water	air	steel
C	steel	air	water
D	water	steel	air

14 · Part 3 Supplement Sound [1]

A fishing boat sends a short pulse of ultrasound vertically downwards into the sea. It receives an echo from a shoal of fish 0.24 s after sending the pulse, and an echo from the seabed 0.40 s after sending the pulse.

The speed of ultrasound in sea water is 1500 m / s.

How far below the shoal of fish is the seabed?

A 120 m

B 240 m

C 180 m

D 300 m

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15 · Part 3 Supplement Sound [1]

Which is an example of ultrasound being used for non-destructive testing of materials?

A killing bacteria on food before it is packed

B checking the contents of suitcases at an airport

C detecting cracks inside a metal railway track

D measuring the surface temperature of a hot engine

16 · Part 3 Core Sound [5]

A loudspeaker contains a paper cone. When the loudspeaker is switched on, the cone moves rapidly backwards and forwards and a sound is heard.

(a) Describe how the cone produces a sound wave in the air in front of it. [2]

(b) State the type of wave that sound is, and describe how the air particles move as the sound wave passes. [2]

(c) The loudspeaker produces a sound of frequency 50 Hz. State whether a person with normal hearing can hear this sound. Give a reason for your answer. [1]

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17 · Part 3 Core Sound [5]

A mobile phone is set to ring continuously. Its screen lights up while it rings. The phone is hung by a thin thread inside a sealed glass container. A vacuum pump is connected to the container and slowly pumps the air out.

(a) Describe what a student standing next to the container hears as the air is pumped out. [1]

(b) Explain your answer to (a). [2]

(c) The student can still see the phone's screen light up after the air has been pumped out. State what this shows about light. [1]

(d) Even when the pump has removed as much air as it can, the student can still hear a very faint sound. Suggest one reason for this. [1]

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18 · Part 3 Core Sound [7]

Two students are asked to measure the speed of sound in air outdoors. They are given two wooden blocks, a stopwatch and a long tape measure.

(a) Describe how the students could use this equipment to determine the speed of sound in air. [4]

(b) In one experiment the students stand 180 m apart. They repeat the timing three times and record these times:

0.52 s, 0.55 s, 0.53 s

Calculate the speed of sound in air from these results. [3]

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19 · Part 3 Core Sound [6]

A student stands 66 m in front of the large, flat wall of a school building. She claps her hands at a steady rate. She adjusts the rate until each clap happens at exactly the same moment as she hears the echo of the clap before it.

(a) State what is meant by an echo. [1]

(b) A friend starts a stopwatch on one clap, counting it as clap zero. He stops the stopwatch on clap number 20. The stopwatch reads 8.0 s. Calculate the time between one clap and the next. [1]

(c) Calculate the speed of sound in air given by this experiment. [3]

(d) Suggest why the friend times 20 intervals between claps instead of timing just one interval. [1]

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20 · Part 3 Core Sound [5]

A microphone is connected to an oscilloscope. The oscilloscope settings are not changed during this question.

When a tuning fork P is struck and held near the microphone, the trace shows 4 complete waves across the screen. The peaks of the trace are 2.0 cm above the centre line.

(a) A second sound, Q, is louder than sound P but has the same pitch. State two ways in which the trace for Q compares with the trace for P. [2]

(b) A third sound, R, gives a trace with 8 complete waves across the screen. The peaks are 2.0 cm above the centre line. Compare the pitch and the loudness of sound R with those of sound P. [2]

(c) State the property of a sound wave that determines its pitch. [1]

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21 · Part 3 Core Sound [5]

A hearing-test machine plays sounds through headphones, one at a time, at these frequencies:

15 Hz, 250 Hz, 4000 Hz, 18 000 Hz, 30 000 Hz

(a) State the approximate range of frequencies that a young person with normal hearing can hear. [1]

(b) Identify the two frequencies in the list that a young person with normal hearing cannot hear. [2]

(c) Sound with a frequency of 30 000 Hz has a special name. State this name and define it. [2]

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22 · Part 3 Core Sound [6]

A ship is stationary in thick fog. The captain sounds the ship's horn once and hears an echo from an iceberg 3.0 s later.

The speed of sound in air is 340 m / s.

(a) State how the echo is produced. [1]

(b) Calculate the distance between the ship and the iceberg. [3]

(c) The ship then starts to move slowly towards the iceberg and the captain sounds the horn again. State and explain how the time between sounding the horn and hearing the echo changes. [2]

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23 · Part 3 Core Sound [6]

During a storm, a student sees a flash of lightning and hears the thunder 4.5 s later. The thunder and the lightning are produced at the same moment.

The speed of sound in air is 340 m / s.

(a) Explain why the student sees the lightning before hearing the thunder. [1]

(b) Calculate the distance between the student and the lightning. [2]

(c) Ten minutes later, the student sees another flash of lightning and hears the thunder 3.0 s later. Assume the storm moves directly towards the student. Calculate the average speed of the storm towards the student during these ten minutes. [3]

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24 · Part 3 Core Sound [6]

A student strikes a tuning fork on a rubber block and holds it about 20 cm from her ear. She hears a steady note.

(a) Describe how the energy of the vibrating tuning fork reaches the student's ear. [3]

(b) Water waves on the surface of a pond are transverse. Compare the way the particles move in a sound wave in air with the way the water particles move in a water wave. [2]

(c) The student strikes the tuning fork again and presses the bottom of its handle onto a wooden table top. The note sounds louder. Suggest why. [1]

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25 · Part 3 Core Sound [6]

A violinist plays a note of frequency 440 Hz. The speed of sound in the air in the room is 330 m / s.

(a) The violinist plays the same note again, but louder. State how the sound wave changes. [1]

(b) Calculate the wavelength of the 440 Hz note in the air. [3]

(c) The violinist presses a finger on the string so that a shorter length of string vibrates. The frequency of the note becomes 660 Hz. State how the note heard changes. [1]

(d) Determine the wavelength of the 660 Hz note in the same air. [1]

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26 · Part 3 Core Sound [7]

A student stands in front of the long, flat wall of a factory and makes a short, sharp sound with a clapper board next to a microphone. An electronic timer connected to the microphone measures the time between the sound and its echo from the wall. She repeats this at different distances from the wall.

distance from wall / m	time for echo / s
20	0.12
40	0.24
60	0.35
80	0.47

(a) Describe how the time for the echo changes as the distance from the wall increases. [1]

(b) Use the result for 80 m to calculate the speed of sound in air. [3]

(c) Predict the time for the echo if the student stands 100 m from the wall. Use your answer to (b). [2]

(d) Suggest why the student uses an electronic timer instead of a stopwatch. [1]

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27 · Part 3 Supplement Sound [6]

A loudspeaker cone vibrates and sends a sound wave through the air. The wave is made of compressions and rarefactions.

(a) Describe what is meant by:

(i) a compression [1]

(ii) a rarefaction. [1]

(b) Explain how the movement of the loudspeaker cone produces a compression followed by a rarefaction. [2]

(c) The distance between the centres of two neighbouring compressions is 0.68 m. The speed of sound in air is 340 m / s. Calculate the frequency of the sound. [2]

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28 · Part 3 Supplement Sound [6]

A worker taps one end of a long steel handrail on a bridge once with a hammer. A student stands at the other end of the handrail, 102 m away, with her ear pressed against the steel. She hears two separate sounds: one travels through the steel and the other travels through the air.

The speed of sound in steel is 5100 m / s. The speed of sound in air is 340 m / s.

(a) State which sound reaches the student first. Explain your answer. [2]

(b) Calculate the time interval between the student hearing the two sounds. [3]

(c) State, in general, how the speeds of sound in solids, in liquids and in gases compare. [1]

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29 · Part 3 Supplement Sound [7]

A survey ship uses sonar to measure the depth of water in a harbour. The ship sends a short pulse of ultrasound straight down and detects the echo.

(a) Describe how the sonar system uses the pulse to find the depth of the water. [2]

(b) The speed of ultrasound in the sea water is 1500 m / s. The echo is detected 0.064 s after the pulse is sent. Calculate the depth of the water. [3]

(c) On another day the water is colder, and the speed of ultrasound in it is lower. The depth of the water is the same. State and explain how the time between sending the pulse and detecting the echo changes. [2]

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30 · Part 3 Supplement Sound [7]

A factory checks aluminium blocks for hidden cracks using ultrasound. A probe placed on the top surface of a block sends in short pulses of ultrasound and detects any echoes. Each block is 0.32 m thick. The speed of ultrasound in aluminium is 6400 m / s.

(a) Describe how ultrasound can show that there is a crack inside a block without damaging it. [2]

(b) Calculate the time between a pulse being sent and the echo from the bottom surface of a block with no crack being detected. [2]

(c) In one block an echo is detected 25 μs after the pulse is sent, before the echo from the bottom surface. Calculate the depth of the crack below the top surface. [2]

(d) Ultrasound is also used to scan a fetus in the womb. State one reason why ultrasound is used for this instead of X-rays. [1]

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